

— HACKATHON SUBMISSION · TEAM TITANS

AI Career Twin

Personal Career Readiness & Interview
Agent

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The Problem Students Face

01

“I don't know what skills I'm missing for my target role.”

02

“I have no structured way to prepare for interviews.”

03

“Career counsellors are expensive and not always accessible.”

Meet AI Career Twin

A personal, always-available career readiness and interview practice agent — built for every student.

GAP

Know your skill gaps

MAP

Get a personalised roadmap

INT

Practice mock interviews

65%

Track your readiness progress

The Student Journey

01

Create
Profile

02

Select
Career
Target

03

Skill Gap
Analysis

04

Learning
Roadmap

05

Career
Missions

06

Mock
Interview →
Dashboard

Student Profile

Profile

Name

Rohan Tharwani

Target

Data Scientist

Skills

Python · 4/5 SQL · 3/5

- Enter name, education, skills with proficiency levels
- Add projects and tech stack
- Optional GitHub and LinkedIn URLs
- Supports multiple profiles for different career targets

Skill Gap Analysis



Role match · Data Scientist

Strong Skills

Python
Data Structures
Git

Needs Improvement

SQL
Statistics
Pandas

Missing Skills

Machine Learning
Model Deployment
Cloud Basics

Every skill classified instantly against your target role requirements.

Personalised Learning Roadmap



12-week roadmap

WEEK 1

Strengthen SQL foundations

WEEK 2

Practice statistics with datasets

WEEK 3

Build first ML model

- **Ordered step-by-step plan:** improve weak skills first, then learn missing ones
- **Weekly task bundles** (Week 1, Week 2...)
- **Daily actionable tips** for every skill gap
- **No complex AI** — simple, transparent rules

Career Missions



This week's missions



Practice Python coding challenges



Learn SQL JOINS



Explain one project aloud

- Checkable weekly tasks generated from your skill gaps
- Examples: “Practice Python coding challenges”, “Learn SQL JOINS”
- Mark missions complete as you progress
- Regenerate missions when gaps change

Mock Interview Practice



Question 2 of 6 · Medium

How would you handle missing values in a dataset?

Type your answer...

Practice score · 2 / 3

- Role-specific questions (6 per career role, 4 roles)
- One question at a time with difficulty indicator
- Type your answer and receive instant practice feedback
- Scoring: keyword matching + answer depth (0–3 per question)
- Retry allowed — only first answer counts toward score
- Sessions saved for review on the Dashboard

Basic practice feedback only — not a professional evaluation.

Career Readiness Progress Indicator

65%

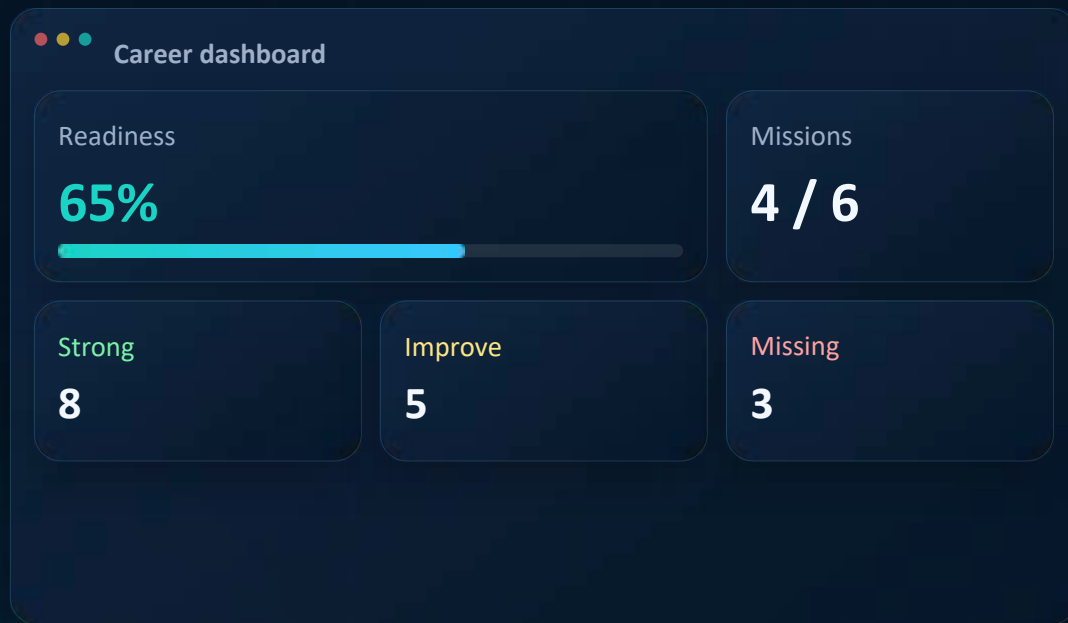


Current career readiness progress



This is a practice progress tracker — it does NOT predict hiring.

Everything in One View



- Profile summary card
- Readiness progress bar + 5 component metrics
- Skill counts: Strong / Needs Improvement / Missing
- Top roadmap steps and daily tips
- Career missions completion tracker
- Interview session history and last score

Technology Used

CORE STACK

Python 3.13 · OOP, type hints, dataclasses

Streamlit · UI framework

JSON · local persistence

pytest · 144 unit tests

IBM BOB

Requirements analysis · Ask mode

Architecture planning · Plan mode

Full code generation · Agent mode

Debugging, testing, and fixing

How IBM Bob Powered This Project

01 · ASK MODE

Analysed problem across 10 dimensions: users, requirements, business rules, exceptions, edge cases

02 · PLAN MODE

Asked 6 clarifying questions, produced 10-sub-task plan with data flow diagram and testing strategy

03 · AGENT MODE

Generated all 8 modules, 14 files, ~1,800 lines of production code

04 · VALIDATE

Ran 144 pytest tests, diagnosed and fixed import errors, verified end-to-end flow

Clean Architecture — 144/144 Tests Pass

career_twin/

- ├─ app.py (Streamlit UI)
- ├─ models.py (Data classes)
- ├─ career_analyzer.py
- ├─ roadmap.py
- ├─ interview.py
- ├─ storage.py
- ├─ exceptions.py
- └─ tests/ (144 tests)

144 / 144

PASSED

test_models.py — 38

test_career_analyzer.py
— 31

test_roadmap.py — 22

test_interview.py — 28

test_storage.py — 25

Summary & What's Next

“AI Career Twin gives every student a structured, honest mirror — showing exactly where they stand and what to do next.”

- ✓ Skill gap analysis for 4 tech careers
- ✓ Personalised roadmap + career missions
- ✓ Mock interview practice with session history

FUTURE SCOPE

Integrate a real LLM for richer interview feedback

Add more career roles and larger question banks

Cloud sync for profile access across devices

github.com/rohantharwani90-prog/AI-Career-Twin

Team Titans | Vivekananda Global University